

NATALIA KONG YI

Product & Service Designer · Experience Strategist · Systems Thinker

Translating research into strategy, and complexity into meaningful experiences

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Complex systems only create value when people can understand, trust, and use them. My work sits at the intersection of research, strategy, and product design, translating complex problems into products, services, and experiences that people can confidently navigate. Across architecture, digital products, and speculative design, I explore the underlying structures that influence behavior, participation, and decision-making. I design products as part of larger systems, where interfaces, services, operations, and human relationships work together to create meaningful experiences.

The projects in this portfolio explore that perspective across different scales—from digital platforms and service ecosystems to hybrid physical–digital infrastructures.

PERSPECTIVES



Enabled by AI, connected devices & health data

Beyond the Clinic. *Conceptual framework used to understand the relationships between patients, knowledge, community, and care services before defining individual interventions.*

DIABETES INTEGRAL HEALTHCARE CENTER

Designing connected care experiences across fragmented healthcare services

Role: Impact Designer · Service Design Team

Year: 2022

Type: Service Design · Healthcare Innovation · Participatory Design

Team: Viernes Peru · Multidisciplinary design team

CHALLENGE

Diabetes management extends beyond medical treatment. The condition reshapes everyday routines, social interactions, and relationships with family and friends, affecting how patients experience and sustain lifestyle changes over time.

The project explored the creation of an integrated healthcare center that combined medical expertise with personalized experiences supporting patients and their social ecosystems.

How can healthcare services support the everyday relationships and practices that influence long-term treatment engagement?

To address this challenge, the team investigated how care could extend beyond clinical interactions, designing new tools, experiences, and service mechanisms that encourage empathy, dialogue, and shared responsibility among patients and their support networks.



Ease
Burden



Share
Experiences



Encourage
Reflection



Actionable
Knowledge



Spark
Curiosity

Learning principles. *Design principles used to guide every learning artifact and experience across the project.*

OUTCOMES

The project resulted in a service ecosystem combining research insights, interaction tools and strategic frameworks to support long-term diabetes care.

Care Ecosystem

Developed a broader understanding of diabetes management by considering the relationships, routines, and social environments that influence patients' everyday experiences.

Shared Practices

Designed and prototyped interaction tools—including conversation cards and an interactive installation—to support empathy, dialogue, and collaborative care between patients and their support networks. MVP tested with 10 patients.

Service Strategy

Created a playbook to guide the development and evaluation of future interaction tools, translating research insights into repeatable approaches for healthcare experiences.

Learning Artifact Design Canvas

Subject: What topic are we exploring? What core idea do we want to develop? How does this idea contribute to the learning and community ecosystem?  1	Objectives: How does this topic support the strategic goals of the healthcare center? What outcomes are we aiming to achieve? Who is this experience designed for? What insights or outputs do we hope to generate?  2	Design & Fabrication: How will this artifact be designed? What are its key components? What format will it take? Does it require any supporting elements for implementation (furniture, lighting or acoustic systems, audiovisual equipment, or other resources)?  7
Relational system: How does this artifact connect with the broader learning ecosystem? What other learning artifacts or community activities does it relate to? What upcoming community events should it support or build upon?  4	Approach: How should this topic be approached? In what context will this artifact be experienced (public, semi-private, or private)? What level of participation or visibility should it encourage?  3	
Principles: How should the learning principles be reflected in this artifact? Rate the relevance of each principle and indicate its importance using the corresponding scale.  5	Users: Who is this artifact designed for? What experience, behavior, or outcome do we want to encourage for each audience?  6	

Learning Artifact Design Canvas. *Framework for translating strategic objectives into cohesive learning experiences.*

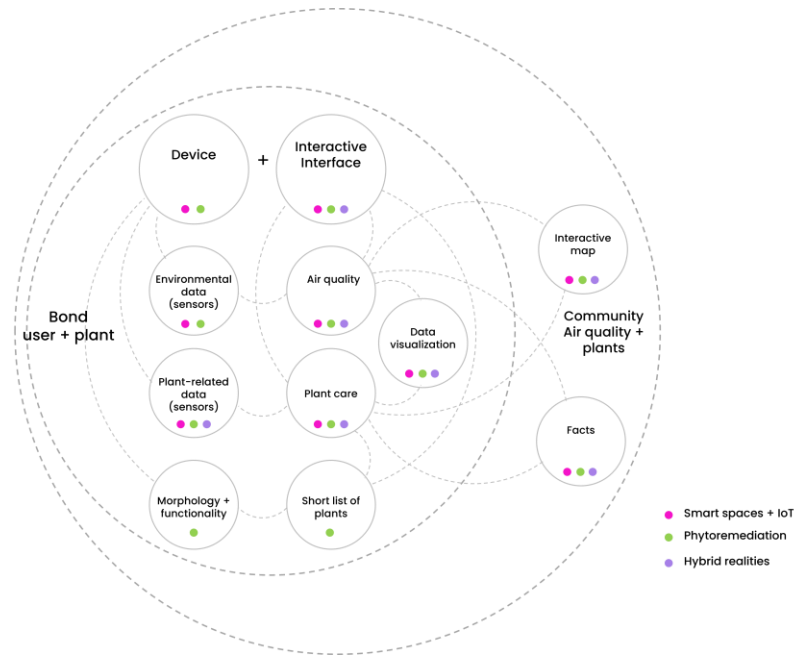
TAKEAWAYS

- Complex challenges require ecosystem perspectives. Understanding healthcare experiences requires looking beyond individual users and considering the relationships, routines, and environments that shape behavior.
- Relationships can become part of the care system. Designing moments of connection and reflection can influence how people understand, manage, and adapt to long-term conditions.
- Design frameworks enable continuity beyond individual interventions. Codifying research insights into reusable approaches allows teams to create, test, and evolve new service interventions over time.

Methods: 6 stakeholder interviews · Service ecosystem mapping · Co-design

workshops · MVP validated with 10 patients

Artifacts: Ecosystem map · Learning Playbook · Family Playbook · Design canvases



Conceptual system map. *Relationships between environmental sensing, digital representation, and user interaction, used to align the platform's architectural logic with the intended user experience.*

HYPERPLANT

Designing hybrid experiences that foster long-term human–plant relationships

Role: Interaction & Systems Designer · Creative Technologist

Type: Master's Thesis · Research through Design

Year: 2024

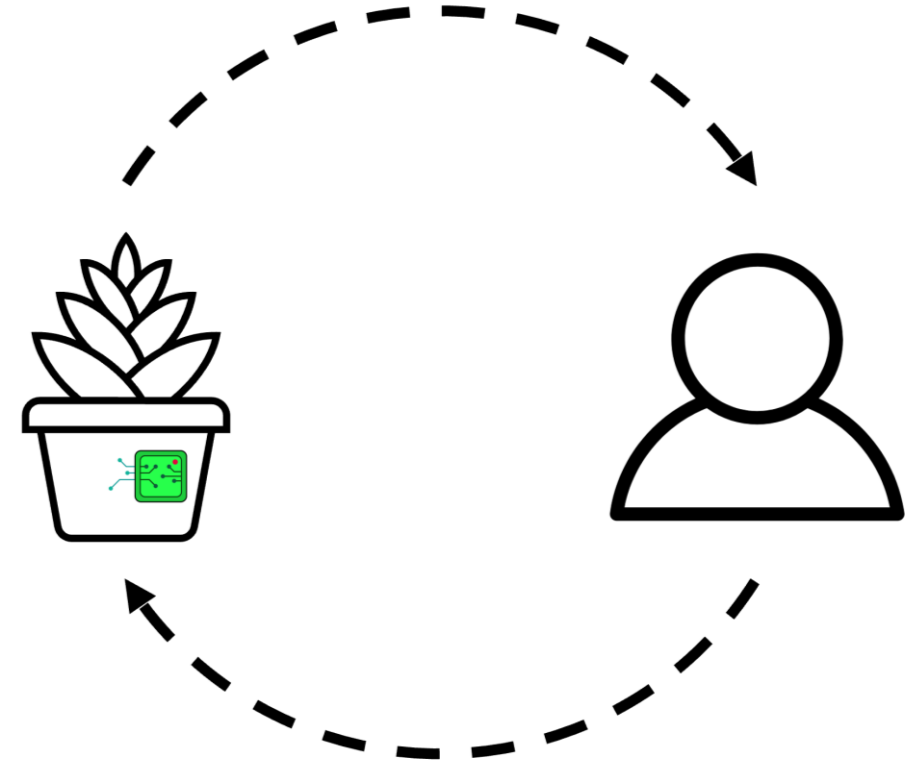
Team: Individual Project

CHALLENGE

Current models of production and consumption often reinforce a cultural separation between humans and nature, making environmental processes difficult to perceive through everyday life. While data can reveal these dynamics, it rarely fosters meaningful awareness or lasting behavioral change.

How might hybrid interactive systems foster awareness, care, and responsibility toward non-human living systems?

To explore this question, I conducted a comparative analysis of bio-inspired interactions and existing approaches to human–plant relationships, identifying recurring patterns and design opportunities. The research process informed the development of a hybrid physical–digital ecosystem combining environmental sensing, living digital twins, and narrative feedback. Rather than visualizing environmental data alone, the system transformed invisible ecological processes into everyday interactions, exploring how technology could extend human perception and create more empathetic relationships with living systems.



Conceptual Framework. Defining the platform architecture and the relationship between sensing and user interaction.

OUTCOMES

The project resulted in a hybrid interaction ecosystem integrating physical computing, digital interfaces, and behavioral evaluation to explore how technology can foster long-term relationships between people and living systems.

Hybrid Interaction Ecosystem

Designed a connected physical–digital ecosystem integrating environmental sensing, interactive devices, and a web platform.

Living Digital Twins

Created expressive digital representations translating environmental data into understandable everyday interactions.

Smart Plant Wearable

Developed a 3D-printed sensing device supporting environmental monitoring and air phytoremediation.

Community Platform

Designed a digital experience for monitoring, storytelling, and future community participation.

Behavioral Evaluation

Designed and conducted a one-week user evaluation using participant diaries and the Octalysis framework.



System Components. The underlying components required to capture, process, and transmit environmental data.

TAKEAWAYS

- Reveal Invisible Systems. Data and interaction can make imperceptible environmental processes understandable through everyday experiences.
- Design for Reciprocity. Making mutual influence visible can strengthen care between humans and non-human living systems.
- Bridge Physical & Digital. Hybrid experiences become more meaningful when digital interactions remain grounded in physical reality.
- Foster Emotional Engagement. Narrative interaction can transform environmental awareness into long-term attachment.
- Future interactions emerge through embodied technologies. The integration of biological systems, sensors, and computation opens new possibilities for how humans perceive and interact with the world around them.

Methods: Research-through-design · Comparative research · Physical prototyping ·

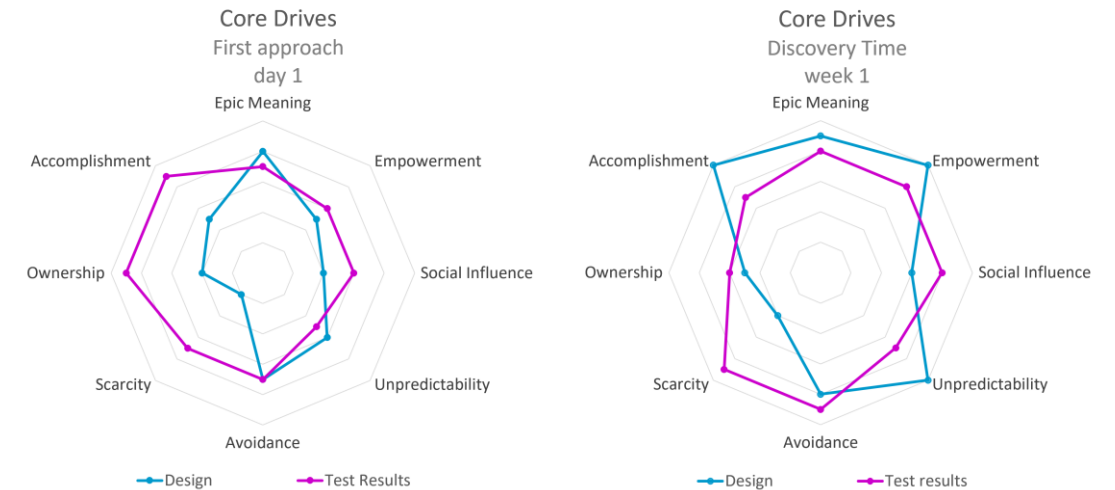
AI-assisted coding · one-week user evaluation · Octalysis analysis

Tools: Arduino · Sensors · Physical computing · 3D printing · Interactive prototyping

Presented at: Chinese Conference on Bio-Inspired Design and Technology (2020) ·

International Image Festival (Colombia, 2021)

More: [Full case study](#) →



Behavioral Analysis. Using the Octalysis framework to measure the alignment between designed motivation models and real user engagement.



Water Flow Series. *Exploring rivers as living data streams, where environmental conditions become visual narratives.*

HYBRID NATURES

Designing participatory infrastructures that reconnect people with living systems through hybrid experiences

Role: Strategy & Systems Designer · Design Research · Creative Technologist

Year: 2025

Type: Master's Thesis (Creative Computing, ELISAVA / UVic-UCC)

Team: Independent project

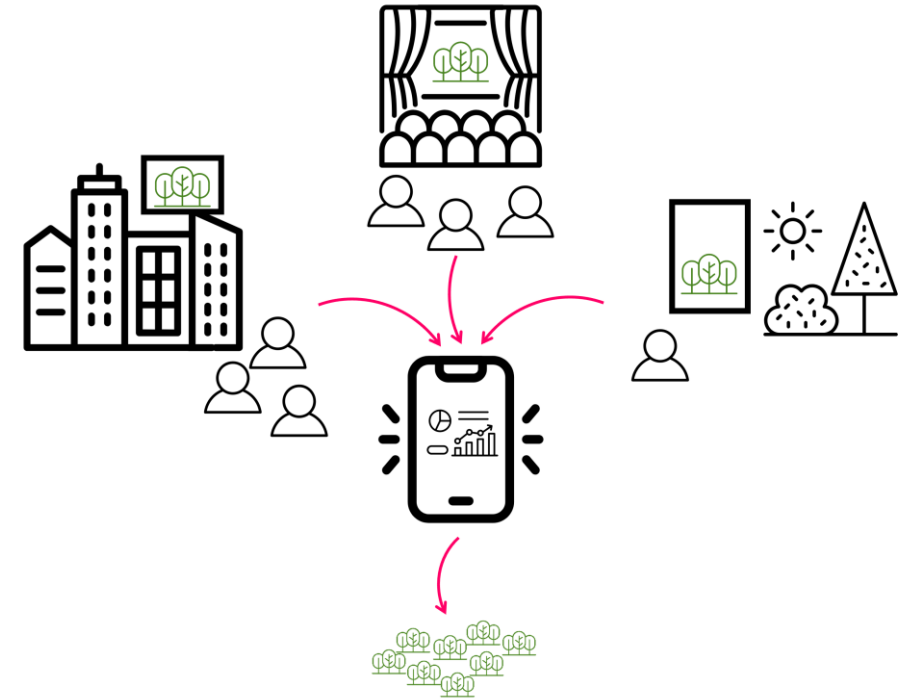
CHALLENGE

Hybrid Natures is a research-through-design project exploring how hybrid physical–digital systems can help people understand, engage, and contribute with complex socio-environmental systems. Rather than treating environmental crises as isolated events, the project explores how data, narrative, and interaction can reveal the relationships between people, territories, and ecological systems.

The first iteration focuses on the Amazon Rainforest as a case study, exploring how distant ecosystems can be made perceptible through participatory digital experiences.

How can design and technology make distant human–nature relationships legible and actionable at a human scale?

Instead of thinking about a standalone product, I approached this project as a systems problem, based on a participatory experience—designing interaction, governance, and contribution as integral parts of the system rather than layers added afterward.



Hybrid Natures Concept. Conceptual framework connecting environmental data, digital ownership, and collective stewardship through a decentralized platform.

OUTCOMES

I developed a proof-of-concept combining interaction design, systems thinking, creative technology, and economic incentives into a digital platform to explore new models of collective environmental stewardship.

Narrative

Designed a data-driven experience guiding users from environmental awareness to active participation.

Participation

Created governance and participation mechanics based on digital ownership and collective decision-making.

Value Model

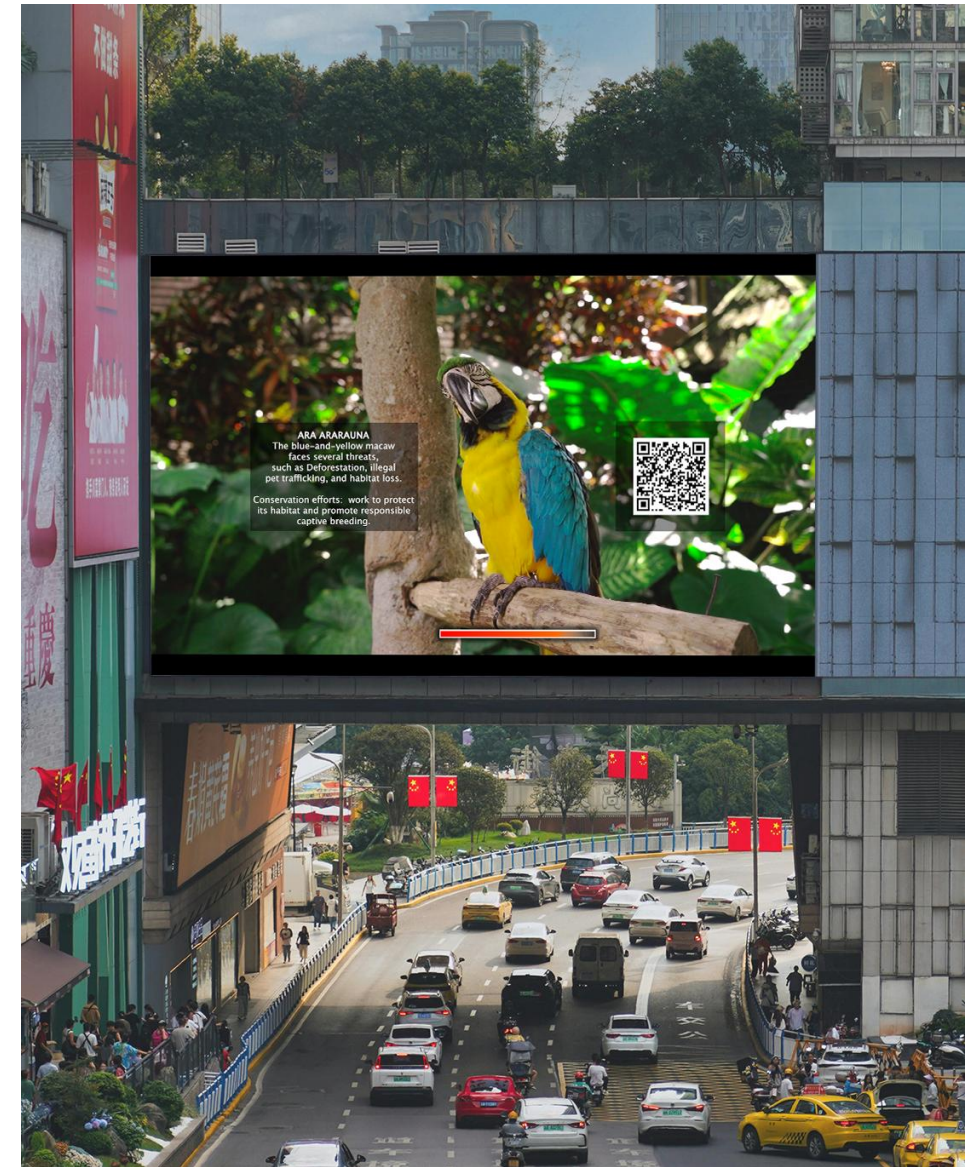
Designed a contribution model allocating 33% of revenue to artists and 67% to conservation NGOs, aligning incentives with conservation goals.

Hybrid Infrastructure

Defined a scalable platform architecture connecting digital interactions with distributed physical touchpoints.

Creative Technology

Developed a functional proof of concept using AI-assisted development, generative visualization, and web technologies.



Decentralized screens (speculative). *Interface concepts for discovering environmental narratives and supporting conservation through digital ownership.*

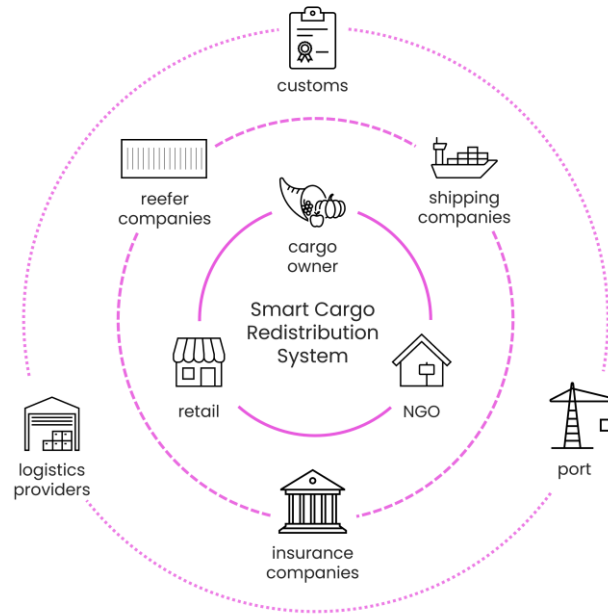
TAKEAWAYS

- Design participation intentionally. Long-term engagement emerges when contribution, ownership, and progression are designed into the experience rather than left to chance.
- Governance shapes the product experience. Decision-making, transparency, and collective authorship influence how people participate—not only how systems operate.
- Narrative makes complexity navigable. Structuring information across multiple scales helps people build mental models of distant and interconnected systems.
- Hybrid systems extend engagement. Physical and digital touchpoints can reinforce one another, creating richer and more persistent participation journeys.
- Speculation is a design method. Building future scenarios makes assumptions tangible, exposing opportunities and informing strategic decisions before implementation.

Tools: p5.js · HTML/CSS · Flourish · AI-assisted development

Presented at: International Colloquium Extractivismos (Argentina, 2025)

More: [Full case study](#) →



Smart Cargo Redistribution System. *Mapping the multi-lateral coordination between maritime IoT data, commercial decision-makers, and humanitarian end-nodes.*

SMART CARGO

Designing resilience into global maritime logistics.

Role: Systems Designer & Strategist

Type: Independent Concept · Systems Strategy

Year: 2026

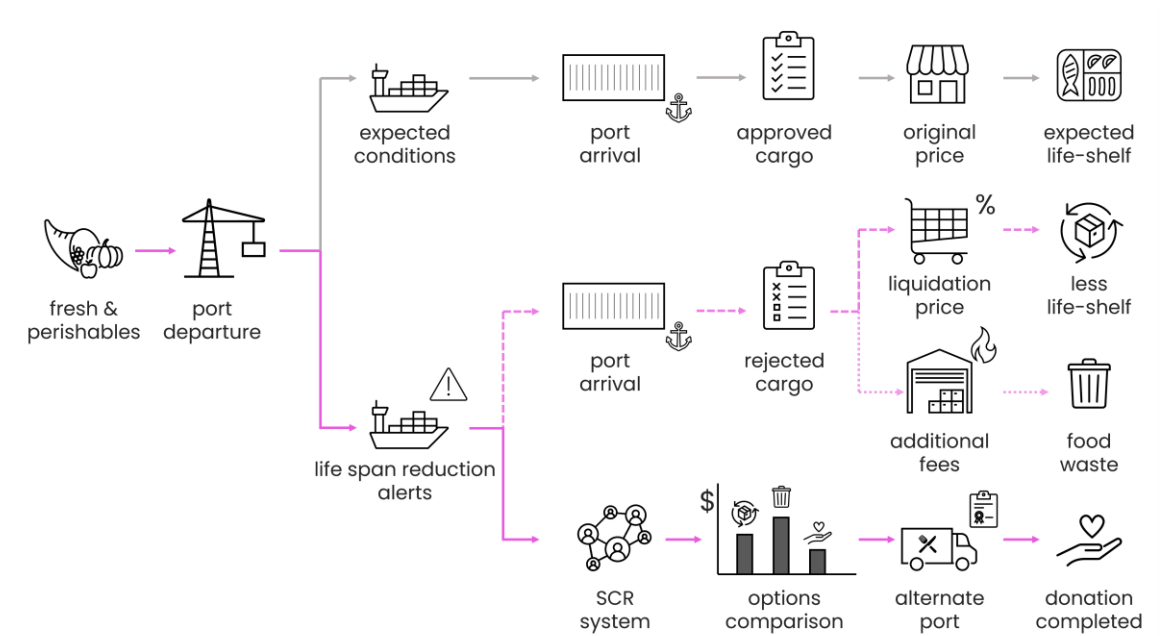
Team: Self-initiated

CHALLENGE

Every year, millions of tons of refrigerated cargo are lost following cold-chain failures, despite much of the food remaining safe for consumption. While existing logistics systems efficiently detect disruptions, they offer limited support for redirecting recoverable cargo toward higher-value social outcomes.

How might we transform logistical failures into humanitarian opportunities by making cargo rescue a more valuable alternative than disposal?

Rather than proposing new logistics infrastructure, SCR explores how existing maritime systems, IoT data, and stakeholder incentives could be orchestrated into a decision-support framework. The concept focuses on interoperability, enabling cargo owners to evaluate redistribution alternatives before food integrity is lost.



Redistribution Logic Journey. *Comparative flow of the intervention journeys and outcomes.*

OUTCOMES

Rescue Decision Framework

A strategic model that reframes cargo recovery as a viable alternative to disposal by comparing operational, economic, and social outcomes.

Redistribution Logic

A systems architecture connecting logistics providers, ports, NGOs, and food banks through coordinated redistribution pathways.

Alternative Engine

A decision-support framework generating multiple rescue scenarios based on cost, timing, and humanitarian impact.

Interoperability Layer

A complementary software strategy designed to integrate with existing maritime infrastructure rather than replace it.

Circular Value Model

A redistribution strategy aligning business incentives with environmental and humanitarian value.

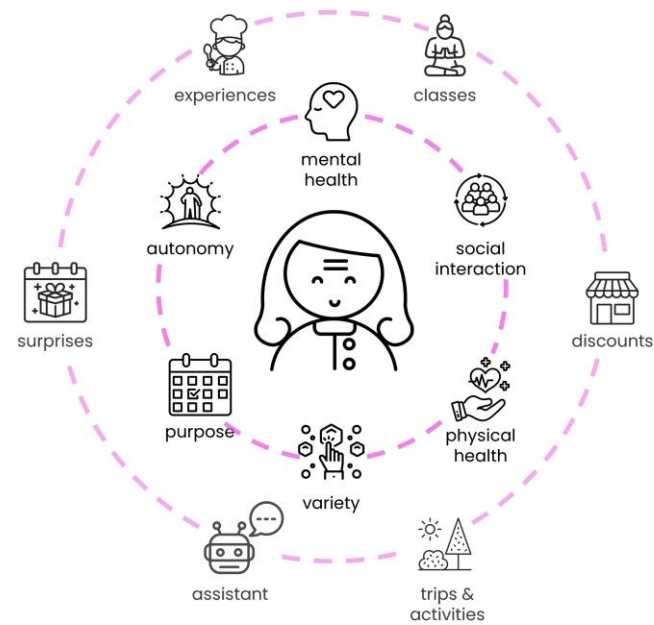
TAKEAWAYS

- Design Incentives. Sustainable systems succeed when the preferred outcome is also the most valuable.
- Orchestrate Systems. Innovation can emerge from connecting existing infrastructure rather than replacing it.
- Support Decisions. Decision frameworks can transform operational failures into strategic opportunities.
- Design for Interoperability. Coordinating fragmented actors creates greater impact than optimizing isolated services.
- Build Resilience. Resilient ecosystems depend on shared information, aligned incentives, and adaptive coordination.

Methods: Systems Thinking · Service Design · Strategic Foresight

Scope: Humanitarian Logistics · Circular Economy · Decision Support Systems

More: [Full case study →](#)



Seniors Club Ecosystem. *Mapping multi-stakeholder participation between seniors, families, and specialized service providers.*

SENIORS CLUB

Designing service ecosystems that strengthen autonomy, belonging, and community in later life

Role: Co-founder · Strategy & Service Designer

Type: Service Strategy & Concept

Year: 2025

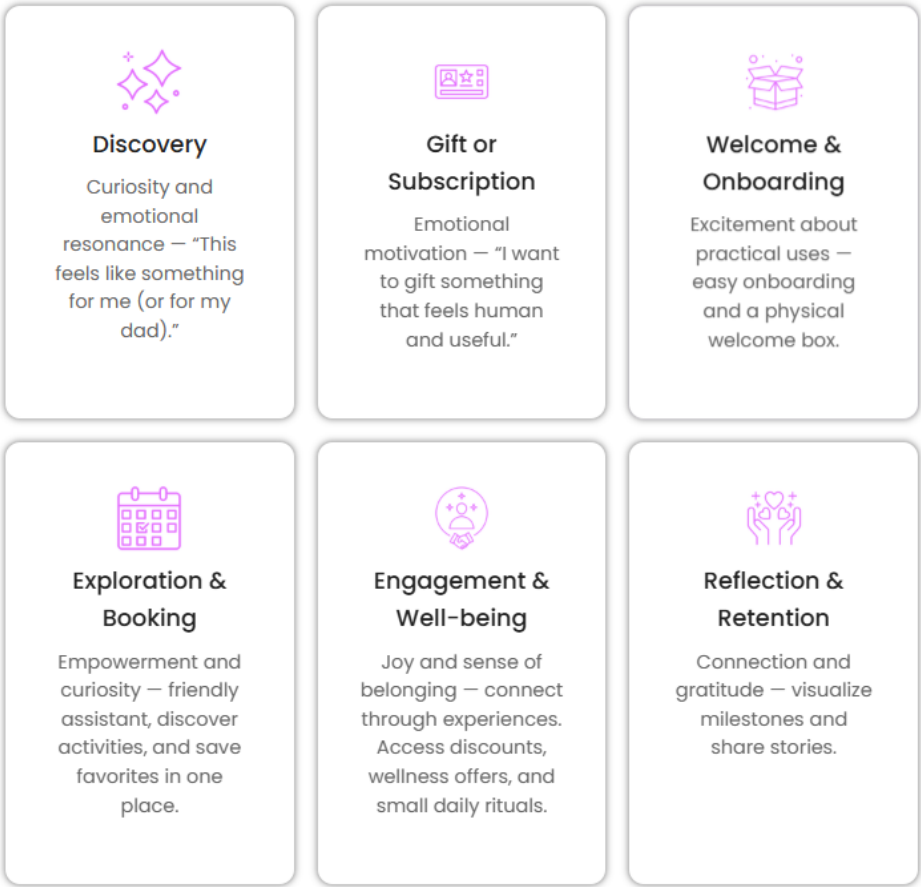
Team: Co-founded initiative

CHALLENGE

As populations age, maintaining independence depends not only on healthcare, but also on opportunities for participation, purpose, and social connection. While many services focus on assistance, fewer explore how design can proactively support active aging through meaningful everyday experiences.

How can a service ecosystem help older adults remain connected, independent, and engaged while enabling families to support them without reducing their autonomy?

To explore this question, we developed a strategic service concept combining research, experience design, and ecosystem thinking. The proposal articulated how curated partnerships, digital coordination, and physical touchpoints could work together to create an ongoing sense of belonging rather than isolated service interactions.



Experience Lifecycle. Defining the emotional and functional touchpoints of the user journey.

OUTCOMES

To explore this opportunity, we developed a strategic service concept integrating ecosystem design, experience architecture, and partnership models to support long-term participation and independent living.

Service Ecosystem

Designed an end-to-end service model connecting seniors, families, and partner organizations through coordinated experiences that encourage participation and long-term engagement.

Experience Framework

Defined the service journey, interaction principles, and touchpoints supporting emotional transitions from discovery and onboarding to participation, reflection, and continued involvement.

Strategic Concept

Established the value proposition, partnership model, and service architecture as a foundation for future prototyping and validation.

TAKEAWAYS

- Care extends beyond healthcare. Supporting well-being requires designing the social relationships and everyday experiences that enable people to remain active and connected.
- Agency encourages participation. Older adults are more likely to engage when services reinforce autonomy and meaningful choice rather than dependence.
- Communities are infrastructures for well-being. Social networks, local organizations, and shared experiences can be designed as interconnected systems supporting long-term quality of life.
- Designing for continuity strengthens belonging. Services that evolve alongside people's lives foster trust, participation, and sustained engagement.
- Strategy shapes experiences before products exist. Clarifying relationships, value exchanges, and ecosystem dynamics provides stronger foundations than designing isolated touchpoints.

Methods: Service blueprinting · User journey mapping · Ecosystem mapping · Value proposition design

Scope: Service strategy · Experience architecture · Partnership model

More: [Full case study](#) →

Education

2024-2025 | Master in Creative Computing. ELISAVA & University of Vic – Central University of Catalonia.

2018-2019 | M. Des in Interactive Design. University of Buenos Aires. (Thesis defended in 2024)

2010 | B. Arch. Pontifical Catholic University of Peru

Research + Exploration Areas

Hybrid Experiences · Systems · Creative Computing · Participatory Models · Generative Design

Publications, Conferences and Lectures

2025 | International Colloquium ExtrActivismos, National University of San Martin. Project presentation (Hybrid Natures).

2022 | Design, Art and Science chair, University of Bogota Jorge Tadeo Lozano. Guest lecture & project presentation (HyperPlant).

2021 | EXPERIENCE DESIGN Korea & Latin America Research Exchange. Human and Design Press. Seoul (KR). Paper co-author (published).

2021 | International Image Festival (CO). Project presentation (HyperPlant)

2020 | Chinese Conference on Bio-inspired Design and Technology (CN). Project presentation (HyperPlant)

Selected Projects

2026 | Kuya Silver Mining Facilities — Architectural Consultant

2026 | Smart Cargo — Strategy Designer

2025 | Seniors Club — Strategy Designer

2025 | Zapallal Mall — Architectural Consultant

2024 | Talara Mall — Architectural Consultant

2024 | HorrorLand — Experience Designer (Immersive Systems)

2023 | Circurela — Designer (Pop-up & Circular Systems)

2022 | Diabetes Integrated Healthcare Center — Experience Designer

2022 | La Molina Mall — Architectural Consultant

2021 | AskMe — Design Director

2019 | S'Dely Corporate Building & Showroom — Principal Architect

2018 | MegaPlaza Resettlement — Architectural Consultant

2017 | Osinergmin — Design Standards Handbooks (National Scale)

2016 | MegaPlaza — Design Standards Handbooks (National Scale)

2015 | VeggieLab — Principal Architect

2014 | Onduline — Marketing & Technical Department Manager

2010 | Participatory Urban Design Workshop — Urban Consultant

BIO